An orbit is continuous falling

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Learning target: Explain a circular orbit with gravity and sideways motion.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 10.*
24 m from the base in the launch direction.

02. Worked example: explain the model
Tangent velocity changes direction; inward force points to the central body throughout.

03. Read and interpret the model
Sideways motion carries it around while gravity curves path; center pull, tangent velocity.

04. Apply the physics
3 orbits; 5 m/s.

05. Explain your reasoning
Gravity supplies the inward net force; continued tangent motion does not need forward thrust. Drag/orbital changes would require corrections.

06. Review: connect an earlier idea *Review: Unit 2, Day 1.*
Changing heading or a curved sequence of positions shows direction changes; velocity includes direction.

07. Exit check
Yes; inward acceleration changes direction.

08. STRETCH: test the model
The 4 m path has twice the circumference and period. Same speed at both radii is not generally a same-body gravitational circular-orbit pair.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 10.*
t=2 s; range 10 m; vertical velocity 20 m/s downward.

02. Worked example: explain the model
Tangent velocity changes direction; inward force points to the central body throughout.

03. Read and interpret the model
Sideways motion carries it around while gravity curves path; center pull, tangent velocity.

04. Apply the physics
v 5 m/s; T=2π16/5=20.1 s; a 25/16=1.56 m/s².

05. Explain your reasoning
Gravity supplies the inward net force; continued tangent motion does not need forward thrust. Drag/orbital changes would require corrections.

06. Review: connect an earlier idea *Review: Unit 2, Day 1.*
Direction changes but speed remains 6 m/s. Instantaneous perpendicular force does no work; velocity is not constant.

07. Exit check
Speed halves; period increases 8 fold because circumference 4×and speed 0.5×.

08. STRETCH: test the model
Speed becomes 1/3, period 27 times, acceleration 1/81 of the originals.